

AI-Enhanced Real-Time Rendering for Laparoscopic Surgical Simulation: Advances and Prospects

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Abstract

This work reviews advances in AI-enhanced real-time rendering for laparoscopic surgical simulation. High-fidelity visualization of tissues and surgical environments has improved training realism, though challenges remain in real-time shadowing and biomechanical accuracy. Artificial intelligence (AI) complements rendering by enabling adaptive learning, objective performance assessment, and personalized feedback. These integrations enhance surgical safety, training efficiency, and scalability. However, high development costs, limited haptic fidelity, and hardware constraints remain obstacles. Future progress depends on mixed reality integration, patient-specific modeling, and collaborative training platforms. This review contributes by consolidating current progress and identifying future directions toward more effective and accessible surgical simulation systems.

CCS Concepts

• **Software and its engineering** → **Virtual worlds training simulations**; • **General and reference** → **Surveys and overviews**.

Keywords

Virtual Surgery, Artificial intelligence, Medical Training, Real-time Rendering

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1 Introduction

Over the past two decades, computer simulation technology has become increasingly important in medical education, particularly in surgical training. Laparoscopic or “keyhole” surgery, introduced in the 1980s, uses 5–10 mm abdominal incisions to perform precise, minimally invasive procedures and has rapidly gained worldwide acceptance. Clinical studies highlight its benefits, Zhu Xiumei et al. reported that laparoscopic gynecological procedures averaged

62.56 minutes compared to 132.55 minutes for laparotomy, with 64% less intraoperative blood loss and a 0% complication rate versus 12% for open surgery [Zhu 2021]. Similar advantages are observed across fields, oncological surgery achieves comparable survival with less pain and better cosmetic results [Kang et al. 2023], hepatobiliary and pancreatic surgery shows shorter hospital stays with equivalent margins [Mishra 2013; Mori et al. 2002], and pediatric laparoscopic appendectomy reduces developmental impact and improves recovery by 30% [Elgabroun and Mishra 2008]. However, the traditional “see one, hand over one, do one” apprenticeship model has become inadequate. Novices make over four times more errors than experts [IJgosse et al. 2020], while limited operating room time and instructor attention restrict opportunities for systematic, personalized feedback [Clavien et al. 2005].

How to overcome the limitations of traditional surgical training? Virtual surgical simulation has emerged as an effective supplement to traditional training methods, offering a safe and highly scalable pathway. Especially virtual reality (VR), which is a computer-generated technology that creates immersive, interactive 3D simulated environments, allowing users to experience a strong sense of presence as if they were in a real-world setting [Alaker et al. 2016]. Within this risk-free virtual space, medical students and residents can repeatedly practice fundamental laparoscopic skills. A study demonstrated that surgeons who reached proficiency through VR simulator training (such as with MIST-VR) subsequently performed laparoscopic cholecystectomies 29% faster and made six times fewer errors in real operations, along with significantly reduced risks of intraoperative hesitation and tissue injury [Seymour et al. 2002].

One of the key factors to virtual surgical simulation being truly effective is how immersive and interactive it can provide. This is where the core value of real-time rendering technology lies. It is crucial for surgical trainees to receive immediate and seamless feedback within virtual environments that accurately represent real-world operations. Early simulators, such as those developed by Surgical Science [Li and Wang 2021], offered realistic graphics and tissue interactions that enhanced learning in ways traditional video instruction could not. Nevertheless, significant technical challenges remain, particularly in balancing physical realism with the stringent requirements of real-time performance. Achieving precise tissue deformation, lighting, and shadowing often demands computational resources beyond the capacity of interactive systems. Recent advances in artificial intelligence (AI) have begun to redefine this long-standing trade-off. Through data-driven modeling, learning-based optimization, and intelligent rendering techniques, AI is transforming the balance between realism and efficiency, offering new possibilities for the evolution of virtual surgical simulation.

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By 2025, real-time rendering technologies have been increasingly driven by artificial intelligence (AI), significantly advancing both the visual realism and educational effectiveness of laparoscopic training systems. Despite these developments, research in this field remains fragmented, with progress scattered across computer graphics, surgical education, and AI-based simulation. A review is therefore warranted to synthesize the current state of the art, identify existing limitations, and outline future research directions.

This work provides an overview of advancements in laparoscopic virtual surgery education, as shown in Figure 1. Specifically, we examine progress in two key areas: real-time rendering technology, encompassing GPU-accelerated 3D visualization, tactile feedback, and immersive interactive systems, and AI technology, which integrates rendering with personalized training feedback. We systematically review the technical foundations and practical implementations, assess the potential and constraints of clinical translation, and outline future directions, emphasizing the fusion of multi-modal technologies and the establishment of standardized evaluation systems.

2 Real-time Rendering

2.1 Research progress and its applications

Current laparoscopic virtual surgery systems require a delicate balance between high-fidelity visual output and low-latency interaction. Conventional methods, such as the Marching Cubes algorithm (1987), achieve basic surface modeling by triangulating medical image data, but struggle to meet the demands of real-time deformation for dynamic organs [Lorensen and Cline 1998]. In 2002, Neyret et al. introduced multi-pass rendering and OpenGL texture extension techniques, achieving real-time rendering at 25 fps on SGI hardware with organ surface textures, specular highlights, and instrument interaction, thereby laying the groundwork for laparoscopic simulators [Neyret et al. 2002]. In a breakthrough in 2024, the EndoGaussian framework incorporated 3D Gaussian Splatting, enabling real-time reconstruction of dynamic endoscopic scenes through spatio-temporal Gaussian tracking and differentiable rendering, significantly improving the fidelity of tissue deformation and lighting details [Liu et al. 2024]. That same year, SplattingAvatar further integrated Gaussian splatting with triangular meshes, achieving real-time rendering exceeding 300 fps for human organs while maintaining compatibility with mainstream animation techniques [Shao et al. 2024].

2.2 Core challenges of real-time rendering

2.2.1 Precise lighting and shadows in narrow cavities. The limited light source in laparoscopic surgery often led to distortion in deep tissue shadows. Such conditions could compromise training effectiveness. Although ElHelw and his team indicated that multi-camera setups and geometric proxies could improve view projection [ElHelw et al. 2004], illumination errors persisted in complex cavities. Furthermore, Qian (2017) confirmed that traditional shading models required physically accurate diffuse and specular reflection mapping, which involved substantial computational costs [Qian et al. 2017].

2.2.2 Realistic tissue deformation and organ interaction. It was challenging to simulate the nonlinear viscoelastic properties of soft tissue in real time, as algorithmic models could not fully reproduce the complex deformation behavior of actual biological tissue [Teixeira and Martins 2023].

2.2.3 Smoke, out-gassing and fluid dynamics simulation. During surgery, numerous complex phenomena occur, such as smoke that obscures vision and fluid interactions requiring precise physical modeling. Deng et al. [Deng 2020a] demonstrated that GPU-based smoke volume rendering enables dynamic obstacle interaction; however, its transfer function still relies on manual adjustment. Silva et al. [Silva et al. 2024] further noted that although the lattice Boltzmann method (LBM) is effective for vortex simulation, it remains difficult to couple with tissue deformation.

2.2.4 Maintain low latency for natural interaction. Low latency is critical in virtual systems to enhance user immersion, particularly for realistic tactile feedback, which requires refresh rates exceeding 300 Hz. McNeely's voxel-based repulsion field achieved force feedback at 1000 Hz [McNeely et al. 2005], though it was limited to rigid objects. Courtecuisse's multi-rate haptic rendering approach (70 fps visual update and 1000 Hz force feedback) addressed this limitation but still encountered computational bottlenecks when exceeding 150 contact points in a liver model [Courtecuisse et al. 2010].

2.3 Countermeasures and technological breakthroughs

2.3.1 Light optimization and shadow reconstruction. To address shadow distortion resulting from complex illumination, lighting optimization is a method worthy of exploration. For example, virtual surgery systems may employ ElHelw's view-projection manifold technique [ElHelw et al. 2004], which dynamically redirects lighting through a geometric proxy representation. This method effectively mitigates cavity shadow deformation and preserves consistent light-surface relationships under varying viewpoints. Consequently, it enhances the photometric accuracy, spatial coherence, and visual realism of surgical scenes, contributing to improved depth perception and user immersion.

2.3.2 3D Gaussian point cloud drives dynamic deformation. The EndoGaussian framework used space-time Gaussian tracking technology to discretize organs into deformable Gaussian units. Differential rendering was used to generate deformation surface and depth maps in real time, and sub-millimeter deformation accuracy (such as liver cutting edge) was achieved while ensuring >30fps. SplattingAvatar further bound Gaussian point clouds to mesh bones and supported dual-mode driving of physics and animation [Liu et al. 2024]. In addition, Qian and his team conducted research on the application of PBR in surgical simulation. While clustering discretized shape-matching regions, but the number of clusters affected overall performance and local distortion, the research team decided to solve this problem by dynamically limiting the tendency to return to target locations [Qian et al. 2017].



Figure 1: Overview of key technologies and review framework.

2.3.3 Fluid-smoke coupling simulation. Integrating Deng's GPU smoke volume rendering and Nie Qing's LBM fluid model, real-time visualization of smoke-tissue interaction could be realized by dynamically adjusting the transfer function through the spatial position of obstacles [Deng 2020b]; during virtual surgery, the pressure distribution of the flow field could be obtained by parallel solution, and then all processors could calculate the regular matrix synchronously. Courtecuisse parallel calculation was introduced to accelerate fluid boundary force feedback [Courtecuisse et al. 2010].

2.3.4 Heterogeneous architecture low latency optimization. Extending McNeely's concept of voxel repulsion field to soft tissue for soft tissue collision simulation could significantly improve computational efficiency by building GPUs, especially for large and complex models. The computing power of GPUs made distance field computation more efficient [McNeely et al. 2005]. Accelerated distance field collision detection; Courtecuisse's "compliant twist + multi-rate loop" architecture was used to separate tactile (1kHz) and visual (90fps) threads, and Jagged Diagonal Storage storage compression improved matrix calculation efficiency.

Real-time rendering technology for laparoscopic virtual surgery in 2025 had shifted from traditional mesh modeling to high-fidelity 3D Gaussian point cloud driving. Frameworks such as EndoGaussian significantly improved dynamic tissue rendering quality through

spatiotemporal Gaussian tracking, but narrow space lighting, multiphase fluid interactions, and millisecond delays remained bottlenecks. In the future, it was necessary to further integrate physical simulation with lightweight AI rendering (such as neural radiation field), and at the same time rely on GPU heterogeneous computing to realize full-domain real-time closed-loop of vision-force-fluid. It was worth emphasizing that the verification of the real surgical environment mentioned above (such as Xiumei Zhu's intraoperative blood loss/complication data) would be the key yardstick for the implementation of technology.

3 AI Technologies for Real-Time Rendering in Laparoscopic Simulation

In medical imaging, accurately determining the camera's pose from endoscopic video footage and performing dense 3D geometric reconstruction of observed anatomical structures are critical tasks in clinical applications. Accuracy pose estimations and high-quality scene reconstructions are indispensable for ensuring accurate surgical navigation and enhancing surgeons' cognitive perception. Dense Simultaneous Localization And Mapping (SLAM) systems present a viable solution for navigation in unfamiliar environments, facilitating precise positioning and detailed environmental mapping concurrently. However, the application of SLAM to endoscopic

scenarios encounters three principal challenges. Firstly, the spatial limitations imposed by the small orifices within the human body constrict the functionality of RGB-D sensors, thereby impeding the acquisition of comprehensive color and depth data. Consequently, the outcomes are typically two-dimensional sparse perspective images, which are inadequate for accurately reconstructing the three-dimensional geometric structure of the scene. Secondly, the confined motion space within the body often circumscribe the range of endoscopic movement, complicating the process of making reliable geometric estimations for areas that are not directly observable. This frequently results in discontinuities within the dense reconstructions. Lastly, numerous existing methodologies are predicated on the presupposition of a static light source, an assumption that does not hold in endoscopic sequences where the light source is in motion, synchronized with the movement of the endoscopic camera, thereby creating a dynamic illumination scenario. These constraints significantly impede the practical application of SLAM in endoscopic imaging sequences[Shan et al. 2024].

4 Challenges

4.1 Data quality constraints on AI development

Research has shown that AI training depends on large-scale, high-quality surgical video datasets, however, obtaining sufficient and reliable data remains challenging [Bodenstedt et al. 2020]. Surgical videos often involve patient privacy concerns and require expert surgeon annotations, leading to limited data volume and inconsistent quality. Consequently, AI models struggle to predict anomalies absent from the training data, directly constraining their accuracy and pedagogical adaptability.

4.2 Contradiction between realism and computational cost

In order to improve the training immersion, the system needed to simulate realistic tissue deformation, bleeding and other effects, but this required powerful computing resources. We found that excessive pursuit of visual and physical realism could lead to system stagnation, soaring hardware costs, and difficulty in popularization in ordinary teaching institutions[Sánchez-Margallo et al. 2014]

4.3 Difficult integration with robotic surgical platforms

Current virtual surgical systems typically operate in isolation and lack deep integration with robotic platforms such as Da Vinci, which are prevalent in clinical practice. As a result, trainees often face difficulties transferring skills acquired in simulators to real robotic procedures, revealing a persistent technical gap [Reza and Bokhari 2024]. Moreover, studies have reported that VR-based training can be less effective than traditional methods for certain programs [Gerson and Van Dam 2003]. Virtual reality training is primarily focused on developing technical rather than decision-making skills. Unlike aviation, where pilots train on highly controlled simulator environments with limited data variability, human anatomy exhibits significant interindividual variation [Heloury et al. 1985], reducing the generalizability of skills learned through a single simulation program [Gerson and Van Dam 2003].

4.4 Ethical controversy over AI-generated content

Scholars pointed out that when teaching AI-generated virtual surgical scenes, images that deviated from real cases might mislead students about anatomical structures or pathological features. This "distorted teaching" had potential ethical risks and required strict review standards[Satapathy et al. 2023].

4.5 High initial and operational costs

Research showed that a high-fidelity virtual laparoscopic training system (such as LAP Mentor) could cost tens of thousands of dollars[Ochs et al. 2024], and maintenance and software updates were expensive. This was a heavy burden on primary hospitals and medical schools.

4.6 Poor accessibility and motion sickness

In some institutions, limited access to equipment, often only a single device required trainees to make reservations, significantly reducing training efficiency. The high cost of such systems further hindered their widespread adoption [Ochs et al. 2024]. Additionally, prolonged use of VR head-mounted displays has been associated with symptoms such as dizziness and nausea [Kelly et al. 2024], which can disrupt training continuity and lead some trainees to discontinue participation prematurely.

These deficiencies are interrelated, funding constraints exacerbate data and hardware limitations, while technical bottlenecks intensify ethical concerns. Addressing these challenges requires coordinated collaboration among industry, academia, and research institutions, alongside the establishment of cost-control mechanisms and ethical standards to ensure both technical feasibility and responsible implementation.

5 Future Directions

5.1 AI-driven personalized surgical rehearsal

In a pilot and study of 14 institutions in Latin America, AI ratings coincided with expert raters' ratings by 93.02%[Belmar et al. 2023], although the technology was not a substitute for expert ratings, but it provided a breakthrough in skills training for physicians in remote areas. However, since the sample size of the algorithm was only 500 and it was not possible to investigate whether it was applicable to other training.

5.2 Mixed Reality (MR) Fusion Training

To address the weakness that reality and modeling were never exactly the same, it might even be possible to import real 3D images into virtual reality software in the near future. This might enable trainees to perform actual operations in virtual environments that looked like real operating fields. In a summary, hybrid training models combining physical simulators and virtual scenes were becoming a new trend. Evidence showed that MR technology superimposed virtual device operations on physical models in real time [Udhayakumar 2024].

5.3 Laparoscopic meta-universe education

AI-driven virtual surgical ecosystems are emerging [Jamshidi et al. 2023]. Studies have shown that platforms integrating digital twins, MR training environments, and real-time assessment systems enable students from different regions to share identical surgical scenarios—for example, African trainees participating in a complex cholecystectomy led by European experts. These platforms can automatically compare operational data with global databases to generate personalized improvement plans.

6 Discussion

Virtual laparoscopic surgery has been reconstructing the training model of surgeons through the fusion of high-fidelity rendering and AI technology. Rendering technology has built realistic surgical scenes, but the core challenges are: first, when lighting is required in the cavity, the rendering of shadows is difficult, and real-time simulation of dynamic tissues such as tissue deformation and blood flow also requires extremely high computing power; second, low-latency rendering is still an important point, but the development of technology has a long way to go.

AI technology plays the role of intelligent coach: on the one hand, it generates personalized error correction suggestions in real time by analyzing the data such as students' operation trajectory and strength; on the other hand, the adaptive system based on reinforcement learning can dynamically adjust the training difficulty and simulate rare complication scenarios. This "data-driven + situational" model significantly shortens the cycle of traditional training that relies on real-life guidance. At the application level, virtual teaching exhibits three values:

- Safety empowerment. VR training has been shown to reduce surgical errors by 6 times and increase operating speed by 29%, especially reducing the risk of accidental injury to critical tissues during surgery.
- Resource optimization. Break through the limitation of time and space, alleviate the shortage of cadaver specimens and uneven resources of tutors, and become the key fulcrum of global surgical education during the epidemic period.
- Evaluation innovation. AI automatic scoring system (such as device path analysis, tissue processing score) replaces subjective evaluation, making assessment more objective.

However, development remains constrained by several realism-related bottlenecks. First, the high cost of equipment limits adoption in grassroots institutions. Second, virtual simulations often fail to replicate the tactile feedback of complex procedures, such as tissue suture tension undermining trainees' confidence in transferring skills to real operations. Looking ahead, progress is needed in three key directions:

- Technology fusion. MR combines reality with modeling, which can improve the sense of presence.
- Personalized deepening. Develop a customized case library by leveraging patient-specific CT and MRI data to simulate individual anatomical variations.
- Ecological construction. Establish global virtual surgery credit mutual recognition system, promote open sharing of resources, and make technology dividends benefit more regions.

Virtual laparoscopic training does not replace traditional clinical practice; rather, it offers a "risk-free trial-and-error → precise feedback → iterative reinforcement" pathway, allowing surgeons to build muscle memory and decision-making experience prior to actual surgery. Although challenges remain, each advancement in rendering optimization and AI algorithms contributes to the foundation of a safer surgical future.

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